

Motivation and introduction: outline I

- The term Information Retrieval (IR)
 - ▶ explanation of terms
 - ▶ example scenario for an IR system
 - ▶ historical view and recent developments
- Tasks for IR systems
- Difference between IR and fact retrieval (search in databases)

Motivation and introduction: outline II

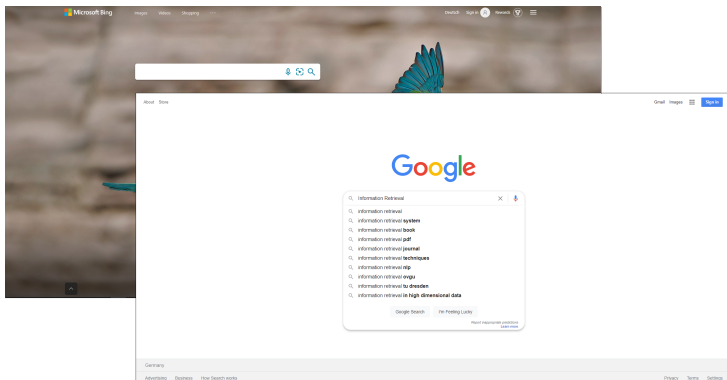
- Basic process of Information Retrieval
 - ▶ Boolean retrieval
 - ▶ Vector space model
 - ▶ Probabilistic Information Retrieval
- Formulation of an information need
- Presentation and evaluation of search results
- Summary and structure of the course

Motivation and introduction

- Topic of Information Retrieval is the **search for documents**
- Specific form of a search can vary:
 - ▶ Which **data set** is being searched?
 - ▶ What **types of documents** are being searched for?
 - ▶ How exactly can the **information need** be formulated?

Examples of IR scenarios

- Search for **text content** in the World Wide Web
- Search for **images** in the World Wide Web
- Search **within** a web presence
- Search in other media types (**video**, **audio**, ...)
- Special filter services (**Google Alerts**,)
- Search in a **library** (one of the roots of IR)



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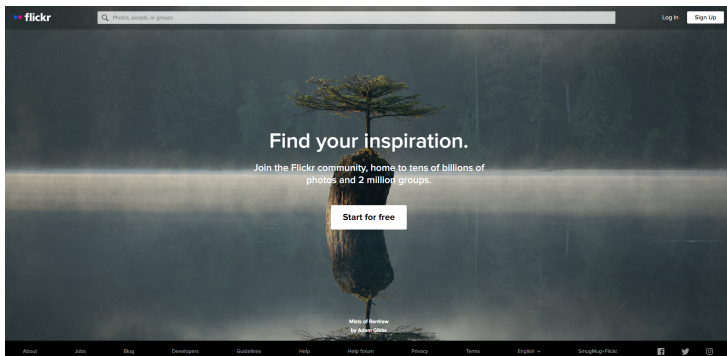
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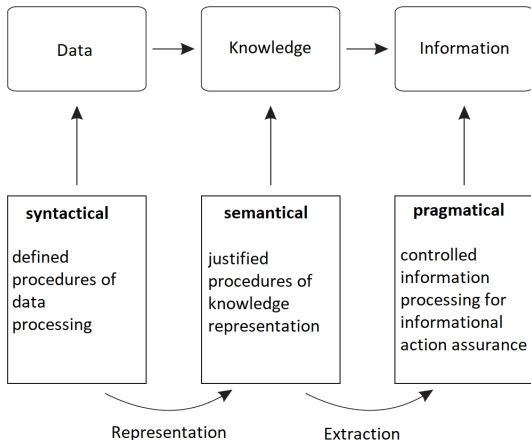
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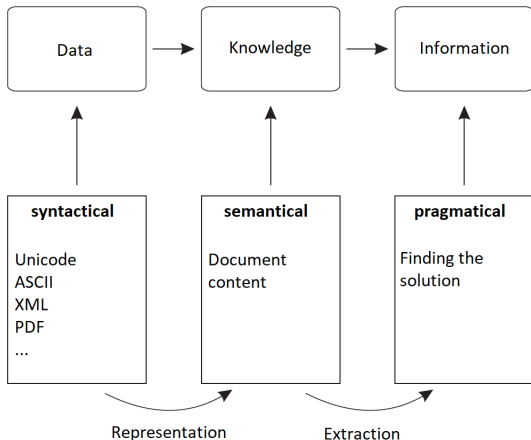
Important aspects of IR

- Formulation of the **information need**
- **Interpretation of the content** of textual and multimedia documents (extraction of semantics)
- **Models** (representation) behind IR systems
- Issues related to the **implementation** of IR systems (such as algorithms and data structures used)
- IR system **assessment** issues

The terms data, knowledge and information in information science (Kuhlen) I



The terms data, knowledge and information in information science (Kuhlen) II



Term “retrieval”:

Merriam-Webster

- locate and bring in
- call to mind again
- get back again
- get and bring back
- recover from storage

Tasks for IR systems

- Classification
 - ▶ catalogs, patent databases, ...
- Query processing
- Browsing
- Information Filtering
- ...

Historical differentiation between DB and IR

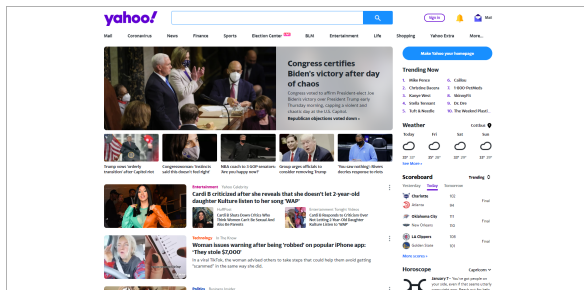
- Database users
 - ▶ programmer
 - ▶ query based on schema
 - ▶ interface is application
- Information Retrieval users
 - ▶ end user
 - ▶ query is vague
 - ▶ direct interaction with the system

Differentiation between Information Retrieval and Fact Retrieval

	Fact Retrieval	Information Retrieval
Matching	Exact Match	Partial Match, Best Match
Inference	Deduction	Induction
Model	Deterministic	Probabilistic
Classification	Monothetic	Polythetic
Query Language	Artificial	Natural
Query specification	Complete	Incomplete
Items wanted	Matching	Relevant
Error response	Sensitive	Insensitive

Search engine functionality

- Search
- Catalog
- ...

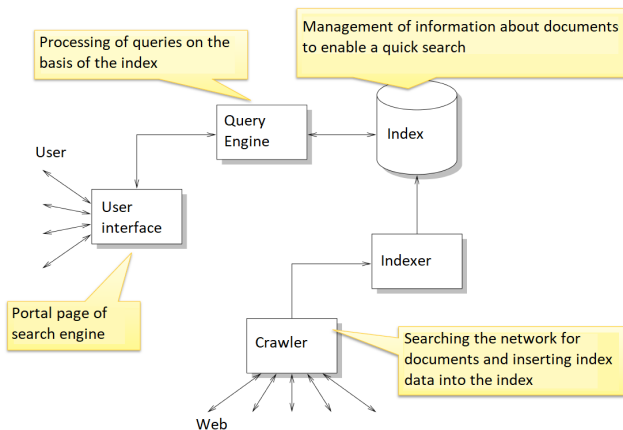


Web catalog functionality



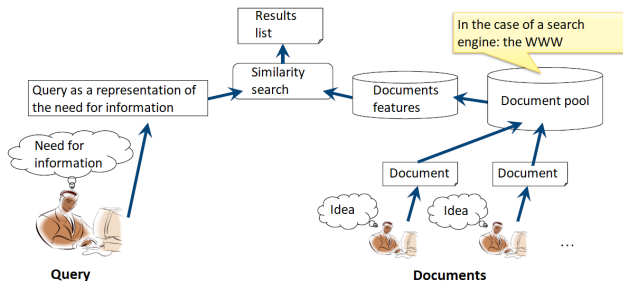
- Manual maintenance
- Historically per hand with support from algorithmic classification

Search engine: what lies behind?



Information Retrieval as a basis

- Information Retrieval (IR) considers information systems in terms of their role in the process of **knowledge transfer from human knowledge producers to information demanders**.



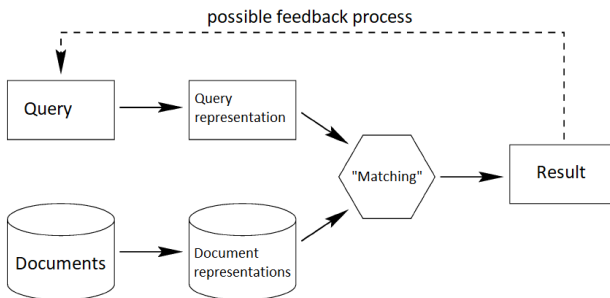
Outline

- ➊ Models of Information Retrieval
 - ➊ The Boolean IR model
 - ➋ The Vector Space Model according to Salton
 - ➌ Quality criteria for IR systems
 - ➍ Consideration of content and quality
 - ➎ Content-based image retrieval (CBIR)
- ➋ Context-based Information Retrieval
- ➌ Search and the market

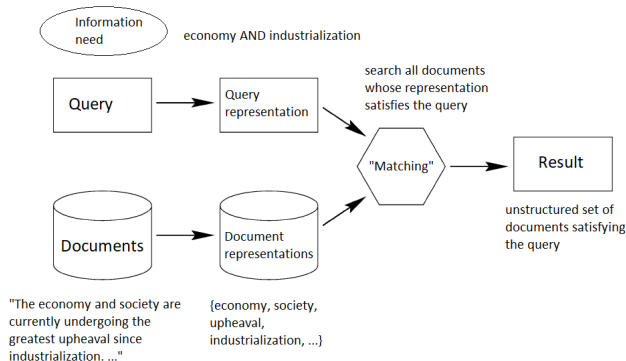
1. Models of Information Retrieval

• Aspects:

- ▶ representation of the **information need**
- ▶ **representation** of the documents
- ▶ **matching**
- ▶ **implementation**



1.1 The Boolean IR Model



Implementation: inverted lists

- Main idea:

- ▶ The **normal representation** is:

the **document** is saved together with all of its words

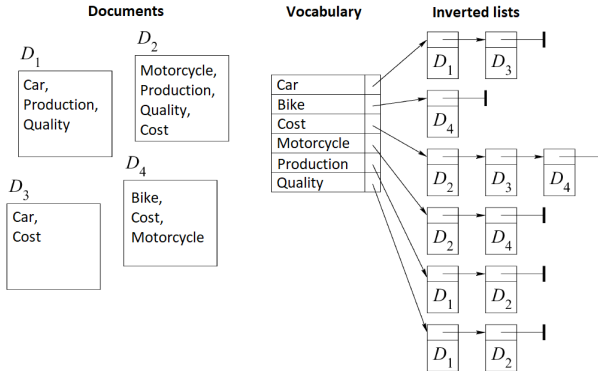
- ▶ However, a **search** does not start with the documents, but **with the words!**
- ▶ So, **invert** the representation:

for each word, save all documents that contain this word

- ▶ **A list of documents containing this word** is managed for each word:

the inverted list

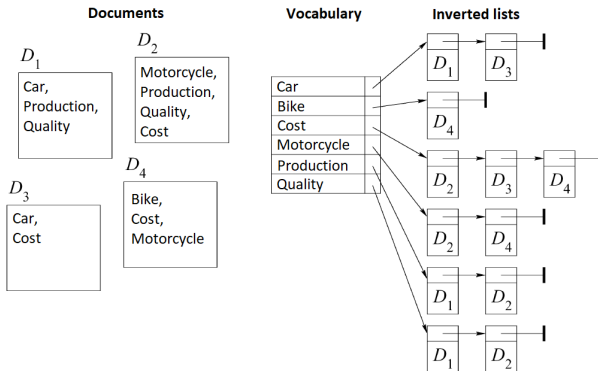
Example: inverted lists



Processing of queries

• Motorcycle AND quality:

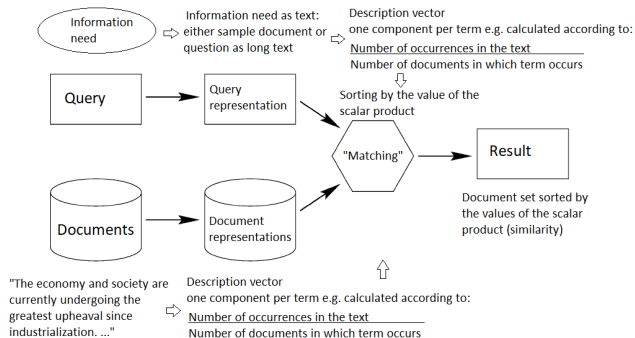
- ▶ **parallel traversing** of the associated lists
- ▶ transfer document references existing in both lists into the **result list**
- ▶ high efficiency due to **sorted** lists



Disadvantages of the Boolean model

- No **stemming** of the words to a basic form
- No **weighting** of words in query and representation
 - ▶ according to the **location** of occurrence
 - ▶ according to the **frequency** of occurrence
- No decomposition of **compound terms**
- **Formulation of a query** is not easy
- Hardly predictable **result size**
- No **ranking** of the result documents

1.2 The Vector Space Model according to Salton



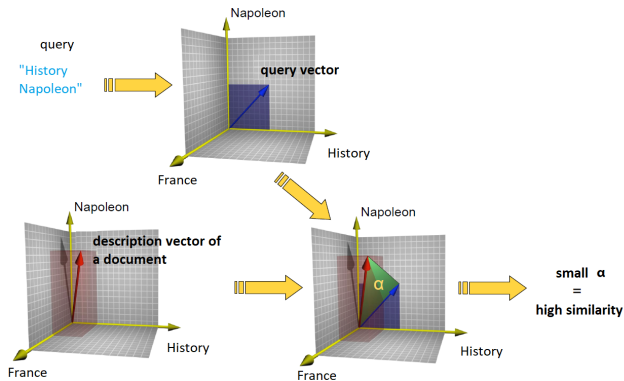
Description vector for a text document

- ① Stop word elimination:
 - ▶ remove words that carry no meaning on their own and occur in almost all texts (and, or, the, ...)
- ② Stemming:
 - ▶ tracing all occurring word forms back to their stem or base (emperors → emperor, ...)
- ③ Synonym/homonym handling:
 - ▶ replace synonyms with preferred form
- ④ Compound words:
 - ▶ break down compound words and/or combine words to compound words
- ⑤ Construction of a vector:
 - ▶ individual components stand for occurrence frequencies of terms (e.g. 1 = history, 2 = Napoleon, 3 = France, ...)

"This is a text about France and Napoleon. It is about revolutions, emperors, wars and also Waterloo. It refers to France and Europe ..."

$\begin{pmatrix} 0 \\ 1 \\ 2 \\ \vdots \\ 0 \\ 1 \\ 0 \\ 1 \end{pmatrix}$

Processing a query



Determination of vector components: The $tf \cdot idf$ formula according to Salton

- t = Number of **terms** in the vocabulary
- N = Number of **documents** in the collection
- n_k = Number of **documents** containing **term k**
- tf_{dk} = **Frequency of occurrence** of term k in document D

The diagram illustrates the components of the $tf \cdot idf$ formula. The formula is presented as $w_{dk} = \frac{tf_{dk} \cdot \log \frac{N}{n_k}}{\sqrt{\sum_{i=1}^t \left(tf_{di} \cdot \log \frac{N}{n_i} \right)^2}}$. Four yellow boxes with arrows point to parts of the formula: 'Consideration of frequency of occurrence' points to tf_{dk} ; 'Consideration of the discriminatory power of the term (idf)' points to $\log \frac{N}{n_k}$; 'Weight of term k in document D' points to the numerator; and 'Normalization to eliminate the influence of the document length' points to the denominator.

$$w_{dk} = \frac{tf_{dk} \cdot \log \frac{N}{n_k}}{\sqrt{\sum_{i=1}^t \left(tf_{di} \cdot \log \frac{N}{n_i} \right)^2}}$$

Consideration of frequency of occurrence

Consideration of the discriminatory power of the term (idf)

Weight of term k in document D

Normalization to eliminate the influence of the document length

1.3 Quality Criteria for IR Systems

- Which aspects are of importance for the evaluation of an IR system?

	IR systems	Database systems
Runtime behavior	important	important
Memory consumption	important	important
Correctness	???	indispensable
Quality of results	very important	covered by correctness

Evaluation of IR systems

- How to measure the **quality of the result**?
- Assumption:
 - ▶ For a given query, it can be **uniquely** decided for each object whether it is **relevant or not**.
- Quality of the result
 - ▶ How **complete** is the result?
 - ★ portion of **relevant documents found in all relevant** documents
 - ★ 50 relevant docs found; 100 relevant docs total $\Rightarrow$ **Recall** = 50%
 - ▶ How **accurate** is the result? Does it contain **only relevant** entries?
 - ★ portion of **relevant documents in the result set**
 - ★ 50 relevant docs found; 75 docs in result $\Rightarrow$ **Precision** = 67%

1.4 Consideration of Content and Quality

- Other criteria:
 - ▶ “quality” of the result documents
 - ▶ type of article (overview article, specific article, ...)
 - ▶ ...
- How to measure the **quality**?
 - ▶ idea (for web engines):
 - ★ Many links pointing to a page indicate quality!
 - ★ an implementation: **PageRank method** from Google

Calculation of page quality

$$R(p) = \varepsilon + (1 - \varepsilon) \cdot \sum_{(q,p)} \frac{R(q)}{\text{outlinks}(q)} \quad (\text{Recursion!})$$

- $R(p)$ = PageRank of page p
- ε = Parameter between 0.1 and 0.2
- (q, p) = Pages q with link to page p
- $\text{outlinks}(q)$ = Number of outbound links from page q
- Values are calculated in an **iterative procedure** of 5 to 50 iterations

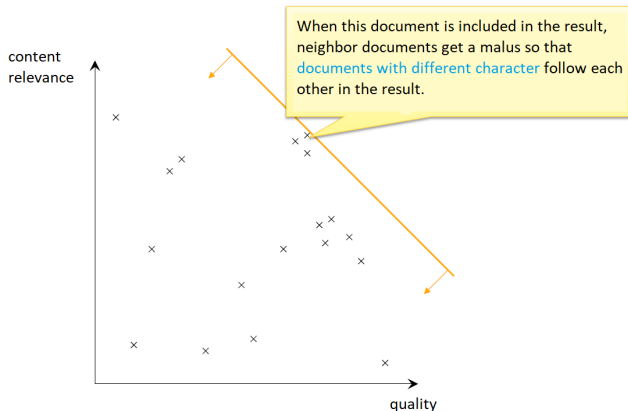
PageRank problems

- There is **no thematic weighting** of the links
 - ▶ links could be weighted according to the thematic relevance of the linking page
- Number of links depends on **“age” of a page**
- There are numerous other **criteria for quality**, e.g.:
 - ▶ up-to-dateness (e.g. via modification date)
 - ▶ availability
 - ▶ information content (number of technical terms/words in the text)
 - ▶ popularity (e.g. via PageRank or social media “likes”)
 - ▶ authority (e.g. via manual assessments)
 - ▶ cohesion (density of content → proximity of the terms used)
- How do you account for **“quality”** and **content relevance**?

Accounting of quality and relevance

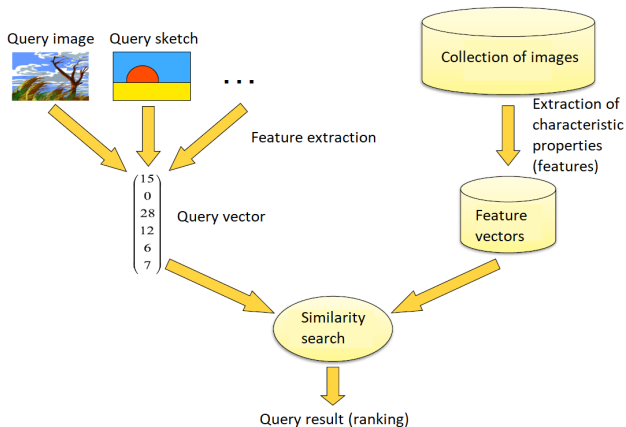
- Addition
- Multiplication
- ...

Approach: mixing the result (addition)



1.5 Content-based Search

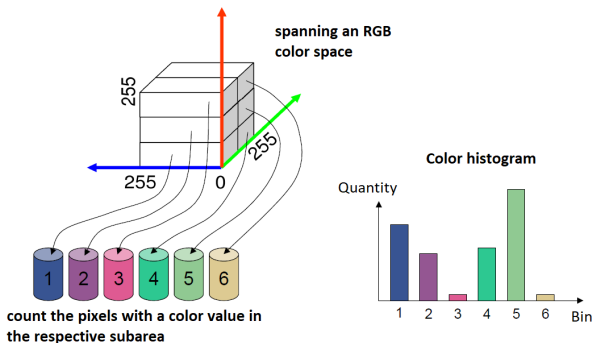
- Transfer to other media



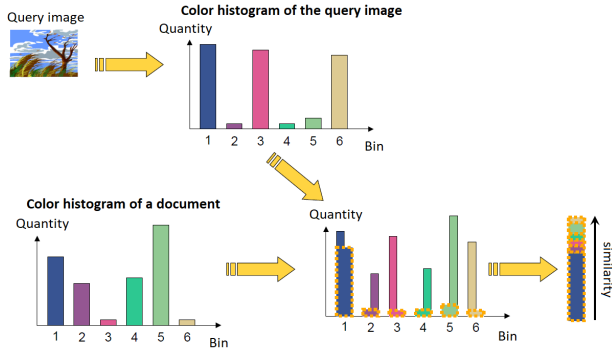
Description vector for images

- Properties:

- ▶ color
- ▶ texture
- ▶ shape



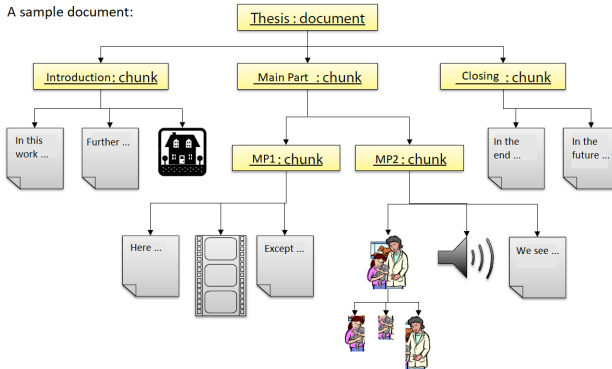
Processing a query



Alternative: object recognition (terms) by CNN

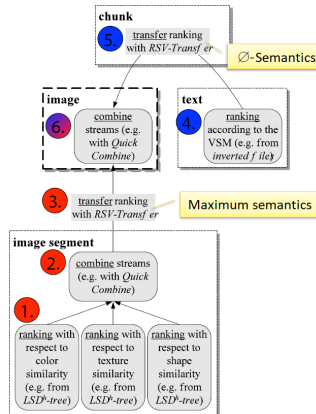
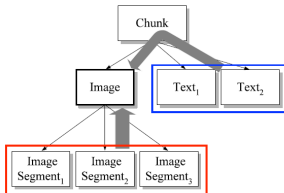
Structured MM documents

A sample document:



Query processing/optimization

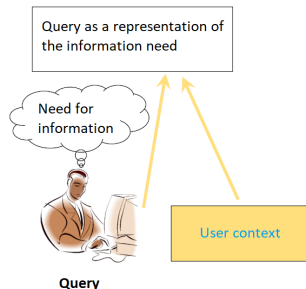
Query: Search all images that contain a given logo and whose text in the "environment" is similar to a given sample text.



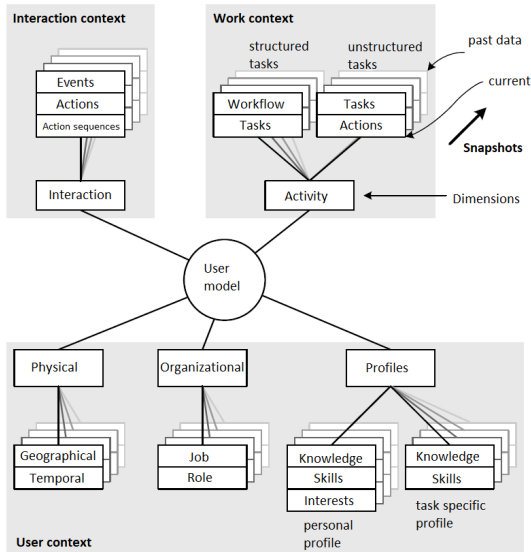
2. Context-based Information Retrieval

- Idea:

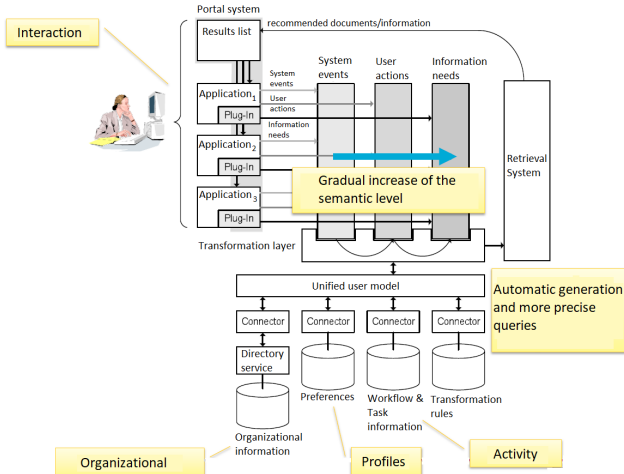
- ▶ Use another source in order to **concretize the information need!**
- ▶ Exploit knowledge from the **context of the user.**
- ▶ This knowledge is used to derive a **more targeted query presentation.**



User model for context information



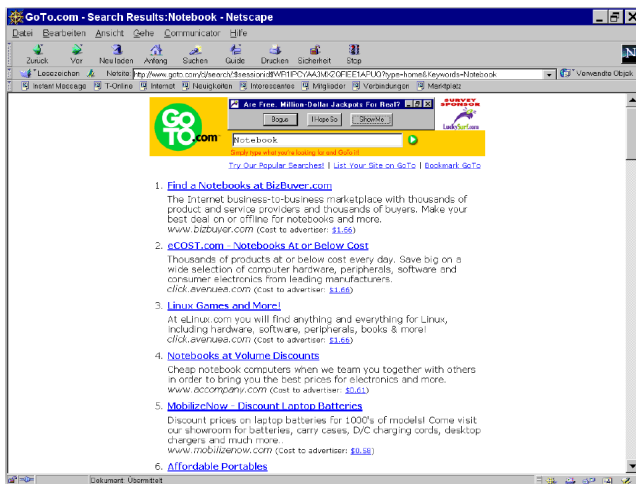
Generic architecture framework



3. Search and the Market

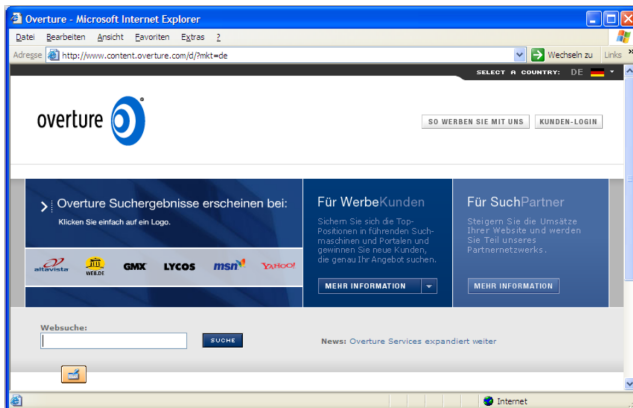
- Idea:
 - ▶ Whoever wants to be found should pay for it!
- Problems:
 - ▶ Whoever can not/will not pay will not be found!

Once upon a time: open payment per click

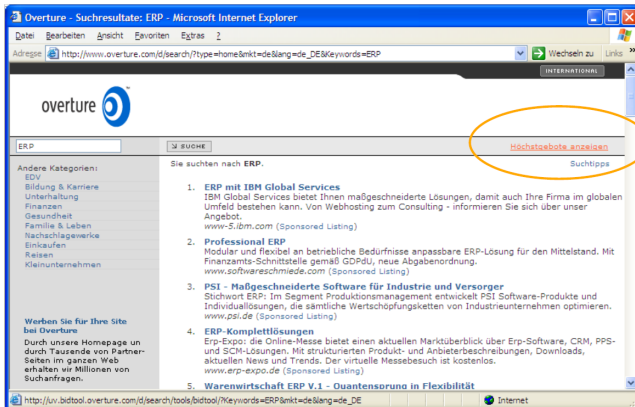




You can still find the info



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